

Characterising Parkinson's Disease-Like Walking Using Wrist-Worn Accelerometers

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Abstract

Changes in walking patterns are a well-known symptom of Parkinson's disease (PD), traditionally assessed through clinical observation. Advances in wearable technology and machine learning (ML) offer new opportunities for early detection, disease monitoring, and treatment evaluation.

This study used wrist-worn accelerometry from the PPMI Verily study, collected from the first seven days of wear, to classify PD status. Raw accelerometer data were segmented into 10-second windows and filtered for active walking using activity recognition models. These walking windows were then classified into labels of healthy control, prodromal PD, or diagnosed PD using several ML models, to determine the best classifier.

A median of 11 hours of active walking data per participant was analysed across 314 individuals: 34 healthy controls, 147 prodromal PD, and 133 diagnosed PD. Using 10-fold group cross-validation, the best-performing was a modified 18-layer ResNet-V2 with self-supervised pre-training, which achieved a mean ($\pm$ standard deviation) macro F1 score of 0.59 ± 0.08 and AUROC of 0.86 ± 0.06 .

These findings highlight the potential of wrist-worn accelerometers powered by deep learning to identify PD-related gait changes. Future work may explore evaluating gait over longer monitoring periods or developing survival models for early risk prediction using an accelerometer-derived gait score.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → **Life and medical sciences**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Keywords

Parkinson's disease, gait, wrist-worn accelerometer, machine learning, activity recognition, deep learning, wearable sensors, digital biomarkers, health monitoring

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1 Introduction

Parkinson's disease (PD) is a neurodegenerative condition characterised by both motor and non-motor changes, including resting tremors, bradykinesia, and altered gait patterns [12, 16]. Treatments like levodopa relieve symptoms by replenishing dopamine, but often lead to dyskinesia and lose effectiveness over time, without addressing the underlying pathology [12, 15, 16]. Wearable technologies and digital biomarkers offer an opportunity to support PD symptoms monitoring. Smartwatches equipped with sensors, such as accelerometers, enable long-term, objective data collection, revealing patterns in motor function and disease progression that are difficult to capture with traditional diagnostic methods like the MDS-UPDRS scale [8, 23]. Deep learning techniques are increasingly used to analyse these datasets and extract meaningful digital biomarkers, showing potential in disease association [1, 18].

Despite these technological advances, the use of wearable data and machine learning (ML) for PD status classification remains largely limited to small-scale studies, often focusing on distinguishing between clearly diagnosed PD patients and healthy controls. It remains unclear whether results can be replicated in larger, more diverse populations. Additionally, little is known about how well these models perform in identifying individuals in the prodromal phase of PD, those who show signs of neurodegeneration but lack definitive motor symptoms of the disease. Furthermore, it remains uncertain which ML model is most effective at distinguishing between different PD statuses.

This research aimed to address these gaps by investigating the classification of PD status using predicted walking behaviour derived from free-living, wrist-worn accelerometer data. The study analysed data from over 300 eligible participants in the PPMI Verily study, employing a variety of ML models to determine which performs best in classifying three distinct PD statuses: diagnosed PD, prodromal PD and healthy controls.



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2 Related Work

2.1 PPMI Verily Dataset

The Parkinson’s Progression Markers Initiative (PPMI) is a longitudinal study launched in 2010 to improve the understanding of PD. It includes clinical, imaging, biologic and genetic data from over 4,000 participants across more than 50 sites worldwide, available at <https://www.ppmi-info.org> [6, 14]. A U.S.-based sub-study, PPMI-Verily, was launched in 2018 and provided participants with a Verily smartwatch to passively collect high-resolution, wrist-worn sensor data over a period of up to two years [1]. Participants wore the device during typical daily activities, resulting in median wear times of 21 hours per day. This sub-study offers a unique dataset of free-living behaviour and motor activity in PD, prodromal, and control cohorts, providing a rich resource for developing digital biomarkers [1, 6, 14].

2.2 Activity and walking recognition

Wearable sensors are essential for activity recognition in health monitoring. Traditional methods include labelling activity intensity using cutpoint thresholds of signal intensity [22]. More recently, ML models have demonstrated superior performance in classifying labels of walking or activity intensity in free-living environments [4, 19, 24]. Using a combination of these models, we can reliably identify periods of active walking in unlabelled data like that captured in the PPMI-Verily study.

2.3 PD status classification

The classification of PD status using wearable sensor data has shown promise in recent studies, although much of the work remains preliminary. Atri et al. (2022) [1] demonstrated the feasibility of using deep learning models to classify PD status based on free-living walking data collected from wrist-worn devices. Their study achieved 100% accuracy in distinguishing between 7 PD and 4 healthy controls, highlighting the potential of such models but also raising concerns regarding scalability and generalisability due to the limited sample size.

Complementing this, Zhang et al. (2020) [26] presented a novel approach using smartphones, participating in the DREAM Parkinson’s Disease Digital Biomarker Challenge. Their method used 3D augmentation of real-world accelerometer data to distinguish PD from control participants, achieving an area under the receiver operating characteristic curve (AUROC) of 0.87. The approach represents a significant improvement in generalisability and robustness over earlier models and serves as a benchmark for future development in mobile health diagnostics.

3 Methods

3.1 Study population

The PPMI dataset included 2,227 participants at the time of data collection. Among these participants, 354 individuals engaged in the Verily sub-study, during which they used the Verily watch for passive monitoring.

Our analysis focused on participants classified into three distinct categories: healthy control, prodromal PD, or diagnosed PD, defined according to the guidelines provided in Marek et al. (2018)

[14]. To ensure the robustness of this analysis, participants labelled as scans without evidence of dopaminergic deficit (SWEDD) or who belonged to the Early Imaging cohort were excluded. These exclusions were made due to the ambiguity surrounding the PD status of such individuals [7]. Furthermore, participants were excluded if their wear time amounted to fewer than 3 days or if any discrepancies or errors were identified in the timestamps of the stored data. Notably, timestamp errors emerged because of attempts made to estimate the unique timestamps associated with each data point attributed to an individual.

3.2 Pre-processing

Pre-processing steps were used to achieve data consistency and comparability across different participants. Firstly, only the initial seven days of monitoring data were used for this analysis. This ensured consistent monitoring periods and aligned with similar studies [10]. In addition, the full monitoring period for many participants coincided with the COVID-19 pandemic, during which activity patterns may have been influenced by lockdown measures. The data was also calibrated to gravity by scaling readings to 1g during stationary periods [21], then filtered using an 8th-order low-pass Butterworth filter with a 20Hz cutoff [3].

3.3 Activity recognition

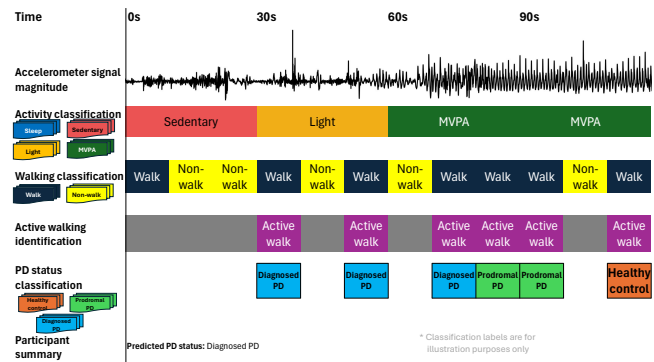


Figure 1: Illustration of ML models used in this analysis.

Human activity recognition was a key component of this analysis, to identify and filter out windows of active walking. To do so, two independent classifiers were used to categorise labels of activity intensity and detect walking behaviour.

The activity classification model, ActiNet (v0.1.0), was based on a modified ResNet-18 [9] architecture, pre-trained using self-supervised learning [25], followed by hidden Markov model (HMM) smoothing [17], and trained on the Capture-24 dataset [4]. This model achieved a mean per-participant macro F1 score of 0.83 ± 0.11 using 5-fold stratified group cross-validation on the training data. To align with the temporal resolution of the walking classifier, the 30-second predictions were resampled into 10-second labels.

The walking classifier also used a modified ResNet-18 [9] architecture, pre-trained using self-supervised learning [25], with final layers trained on data from the OxWalk and MJFF Levodopa Response Study datasets [19, 20]. By incorporating both healthy

individuals and those diagnosed with PD, the model accounted for potential differences in gait patterns between populations [12, 16]. The classifier achieved an average F1 score of 0.91 ± 0.09 across 5-fold group cross-validation on the training data.

To identify periods of active walking, outputs from both classifiers were combined. Specifically, 10-second windows were classified as active walking if they were labelled as walking by the walking classifier and as light or moderate-to-vigorous activity by the activity classifier. All other windows were excluded from subsequent analysis. This is illustrated in Figure 1.

3.4 PD classification

Model Name	Balanced	Other Parameters
Logistic Regression	✓	–
Random Forest	✗	n_estimators: 1000, max_features: sqrt
Balanced Random Forest	✓	n_estimators: 1000, max_features: sqrt, sampling_strategy: not minority
XGBoost	✗	max_depth: 13, n_estimators: 250
ResNet-18	✓	pre-trained: False, learning_rate: 0.0001, patience: 10
SSL ResNet-18	✓	pre-trained: True, learning_rate: 0.0001, patience: 10

Table 1: List of ML models used in this analysis.
SSL = self-supervised learning, sqrt = square root.

Each 10-second active walking window was assigned a ground-truth label corresponding to the PD status of the participant. A series of ML models were then evaluated for their ability to classify PD status using these labelled windows. An overview of the models tested is provided in Table 1.

To address class imbalance in the dataset, most models incorporated balancing strategies, except for the random forest [2] and XGBoost [5] models. For all non-deep learning models, features were extracted from the Euclidean norm of the tri-axial acceleration signal. These hand-crafted features captured statistical, temporal, and spectral signal characteristics.

In addition to traditional models, deep learning approaches using modified 18-layer ResNet-V2 [19] architectures were also assessed. These models operated directly on raw 10-second accelerometer windows and did not require feature engineering. One variant was trained from scratch, while the other was initialised with weights from a self-supervised pre-training task based on temporal arrow-of-time prediction [25]. Both ResNet-18 models employed the Adam optimiser [11] and used inverse weighting cross-entropy loss to mitigate class imbalance [13].

To assess the performance of each model, we used 10-fold stratified group cross-validation, using the same splits for each model. These splits were stratified by class label to ensure a fair distribution of PD status in each fold, necessary due to class imbalance. Data was grouped by participants to ensure no leakage of information between training and testing, offering a fair evaluation of the model's performance on unseen data. The model performance for the test data was aggregated for each fold, evaluated using accuracy, macro F1, AUROC, Cohen's kappa and balanced accuracy.

4 Results

4.1 Study population

The cohort for this analysis comprised a total of 314 participants. Among them, 133 individuals were diagnosed with PD, 147 participants were identified as prodromal, and 34 participants were classified as healthy controls. Figure 2 shows the exclusion flow diagram for participants within this study.

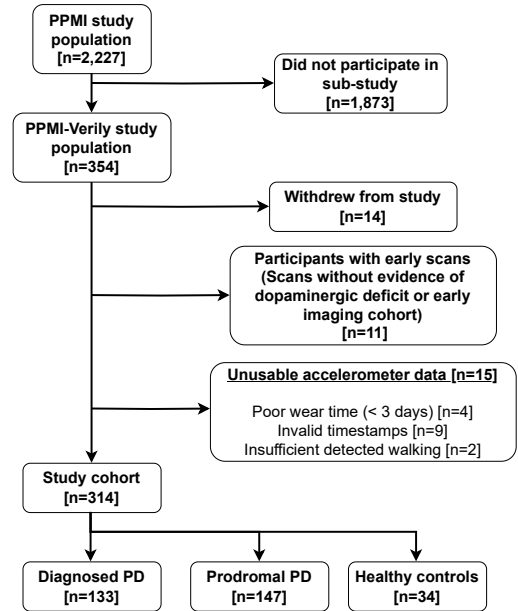


Figure 2: Exclusion flow diagram for study.

4.2 Activity recognition

Figure 3 displays the overall distribution of detected 10-second active walking frame counts from the first 7 days of monitoring for participants. A median of 11 hours of active walking data was detected per participant, ranging between roughly 30 minutes to 25 hours.

4.3 PD classification

The performance of the ML models developed to classify PD status was evaluated per participant in Table 2. Noticeably, the SSL ResNet-18 model outperforms all other models across all metrics evaluated, with a mean ($\pm$ standard deviation) macro F1 score of 0.59 ± 0.08 and AUROC of 0.86 ± 0.06 . The per-participant confusion matrix

Model	Accuracy	Macro F1	AUROC	Cohen's kappa	Balanced Accuracy
Logistic Regression	0.557 ± 0.054	0.475 ± 0.057	0.786 ± 0.038	0.334 ± 0.062	0.550 ± 0.092
Random Forest	0.662 ± 0.056	0.454 ± 0.048	0.812 ± 0.033	0.378 ± 0.104	0.488 ± 0.042
Balanced Random Forest	0.427 ± 0.045	0.376 ± 0.050	0.807 ± 0.038	0.242 ± 0.050	0.529 ± 0.060
XGBoost	0.668 ± 0.056	0.462 ± 0.047	0.813 ± 0.034	0.390 ± 0.104	0.493 ± 0.043
ResNet-18	0.632 ± 0.100	0.521 ± 0.072	0.840 ± 0.052	0.430 ± 0.124	0.579 ± 0.083
SSL ResNet-18	0.685 ± 0.070	0.586 ± 0.082	0.856 ± 0.056	0.501 ± 0.099	0.615 ± 0.108

Table 2: PD status classification performance using active walking windows. Metrics are presented as mean ± standard deviation.

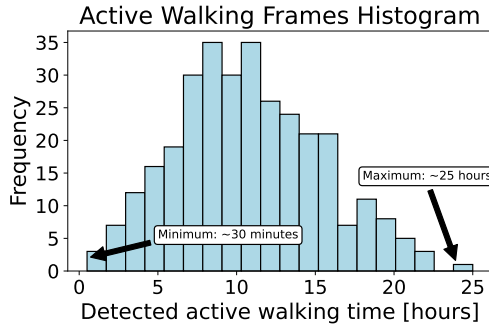


Figure 3: Histogram for the amount of detected active walking windows in the study.

for the best-performing model, shown in Figure 4. This confusion matrix reveals accurate classification of diagnosed PD, but greater confusion between prodromal PD and healthy controls.

		Confusion Matrix		
True Label	Healthy Control	0.382 (13)	0.529 (18)	0.088 (3)
	Prodromal PD	0.327 (48)	0.558 (82)	0.116 (17)
	Diagnosed PD	0.060 (8)	0.038 (5)	0.902 (120)
		Healthy Control	Prodromal PD	Diagnosed PD
		Predicted Label		

Figure 4: Confusion matrix for SSL ResNet-18 model to classify labels of PD status using active walking windows.

5 Discussion

This study investigated the classification of PD status using wrist-worn accelerometer data collected in free-living conditions, focusing on periods of active walking. Among the models tested, the self-supervised ResNet-18 model outperformed all traditional ML approaches, demonstrating a strong ability to capture disease-relevant walking patterns.

Despite this added complexity of three PD status labels, model performance remained comparable to that reported in previous studies [26]. These results highlight the model's capacity to generalise to real-world settings, marking a step forward in digital phenotyping of PD gait for wrist devices.

The study's main strengths lie in the application of robust, modern deep learning techniques and the use of diverse, real-world datasets. However, limitations include a relatively small and imbalanced sample, limited monitoring duration, and a participant-level label aggregation strategy that may not fully capture temporal nuances in symptom expression. Addressing these limitations will be critical in future research to enhance the reliability and granularity of PD classification models.

Future work should focus on expanding participant cohorts, improving class imbalance, validating model findings on the external dataset, and using this PD-like gait score for risk analysis, for early-stage detection of PD.

6 Conclusion

This study demonstrates the potential of deep learning to classify PD status using walking behaviour captured from a wrist-worn sensor during free-living.

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